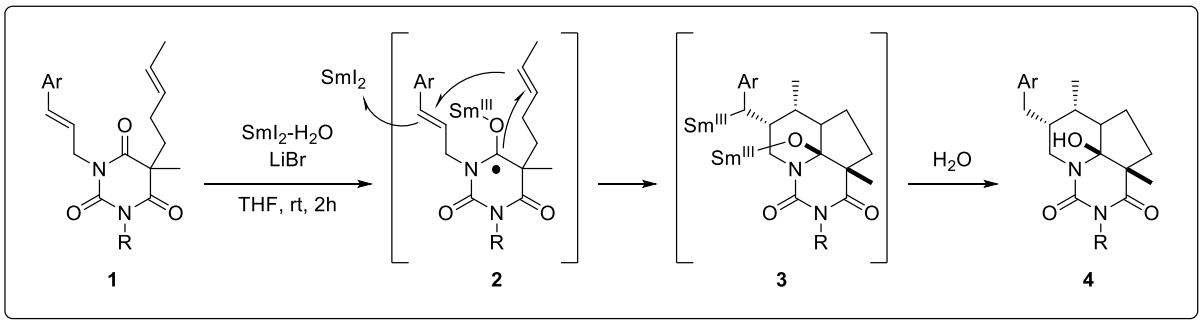
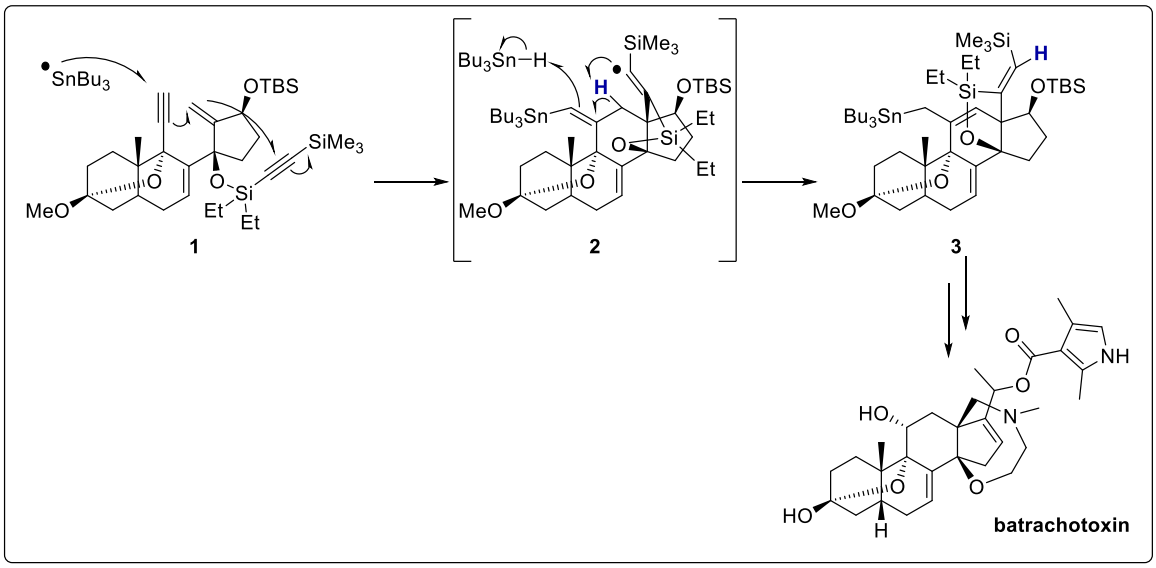
1. **Barbiturate-type substrate 1** was converted to a complex product via a radical cascade reaction. Draw the structures of the intermediates and final product and provide a reaction mechanism.



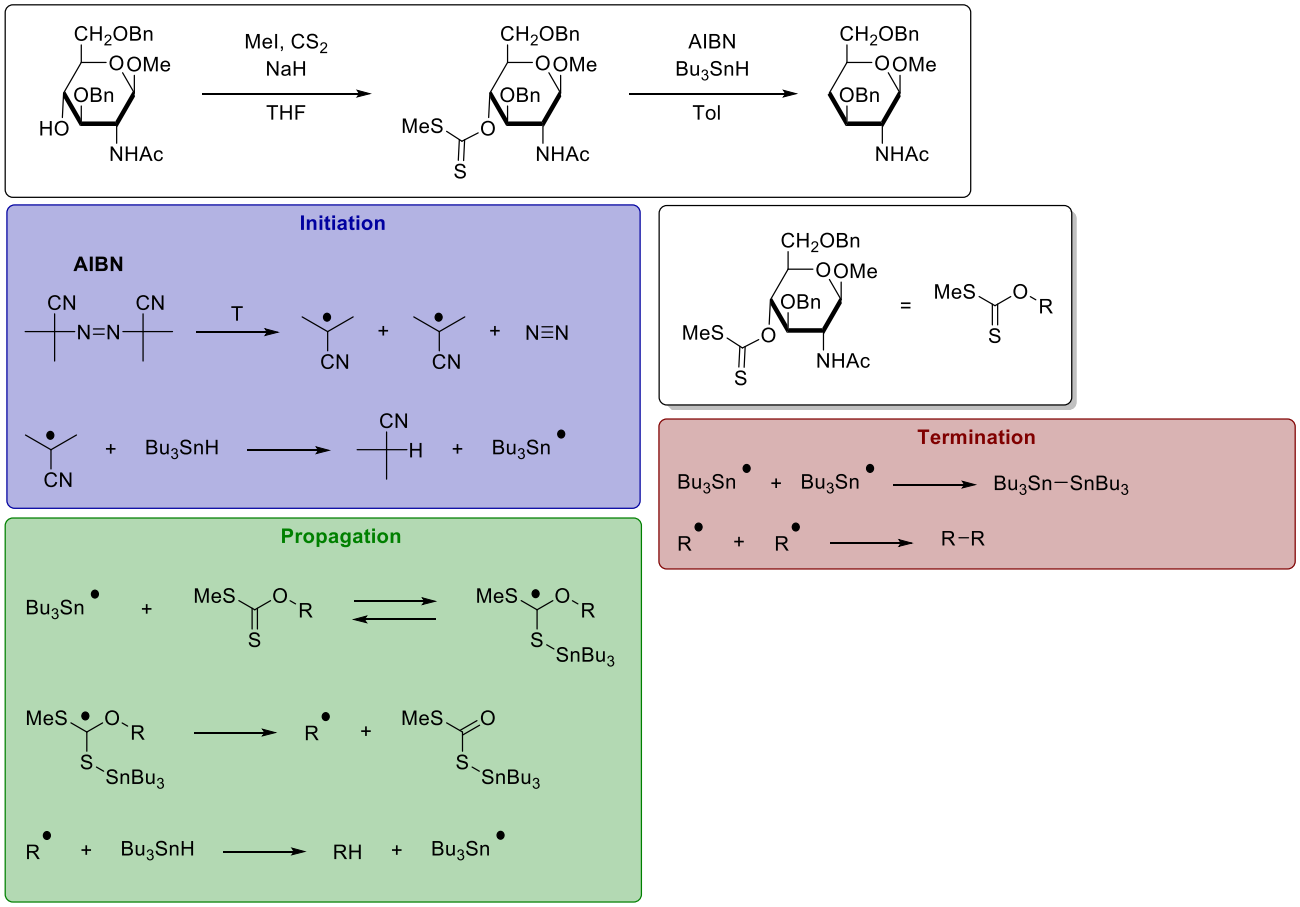
The cascade cyclization reaction shown above was reported by Procter and co-workers (*J. Am. Chem. Soc.* **2016**, 138, 7770-7775). Starting from **compound 1** a ketyl radical is generated in the presence of SmI_2. The **ketyl radical 2** undergoes two successive radical cyclization's to form **3**. Protonation with water then produces the final **compound 4**.

2. **Compound 1** is an advanced intermediate in the total synthesis of **batrachotoxin**. Given the reaction conditions propose a structure for the unstable **intermediate 2** and subsequent **intermediate 3**.



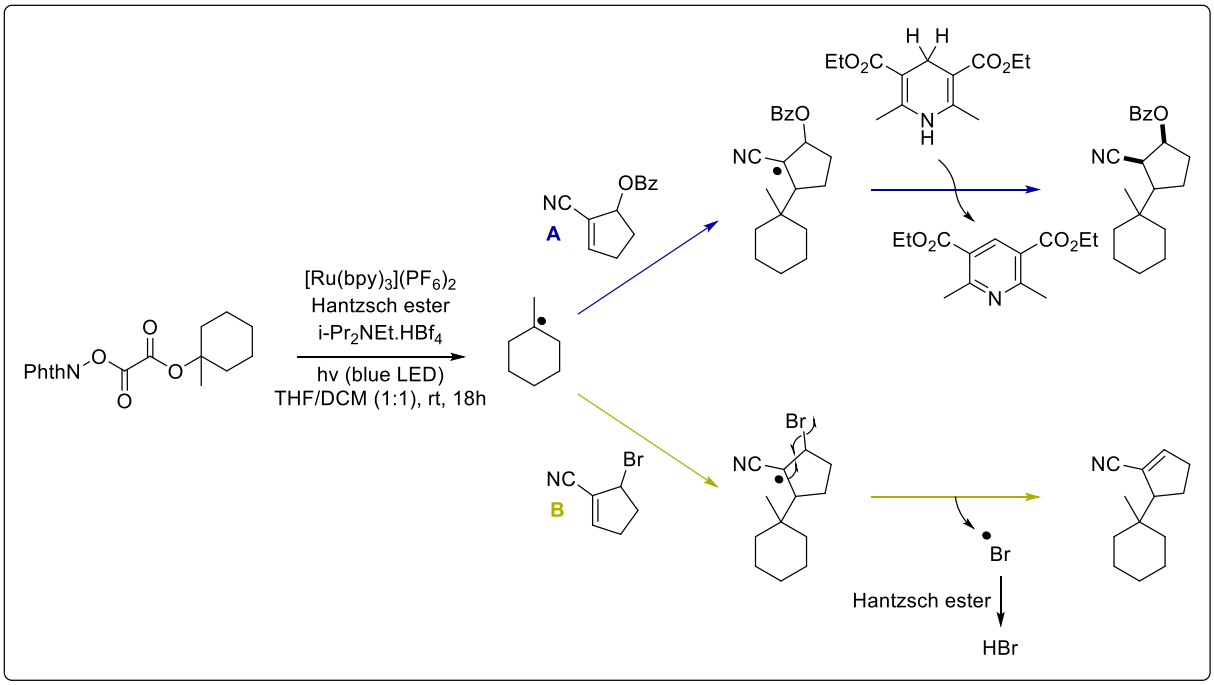
The total synthesis of **batrachotoxin** was reported by Du Bois and co-workers (*Science* **2016**, 354, 865-869). Addition of stannyl radicals on the terminal alkyne of **intermediate 1** is followed by a 6-endo ring-closure and capture of the resulting tertiary radical by the alkynylsilyl fragment resulting in **vinyl radical intermediate 2**. The **vinyl radical intermediate** 2 further undergoes a 1,4-hydrogen atom shift to afford the final **allylic stannane intermediate 3**. This intermediate was further converted to **batrachotoxin** in multiple steps. Note: the 6-endo ring-closure could be the result of a more complex sequence, namely a 5-exo cyclization followed by a 3-exo cyclization and opening of the cyclopropane ring thus formed (see *Tetrahedron Lett.* **1986**, 27, 4525 and *Tetrahedron Lett.* **1986**, 27, 4529).

3. Propose a radical strategy for the replacement of the hydroxyl group by a hydrogen atom? Draw the mechanism of the radical step showing the initiation, propagation, and potential termination steps.



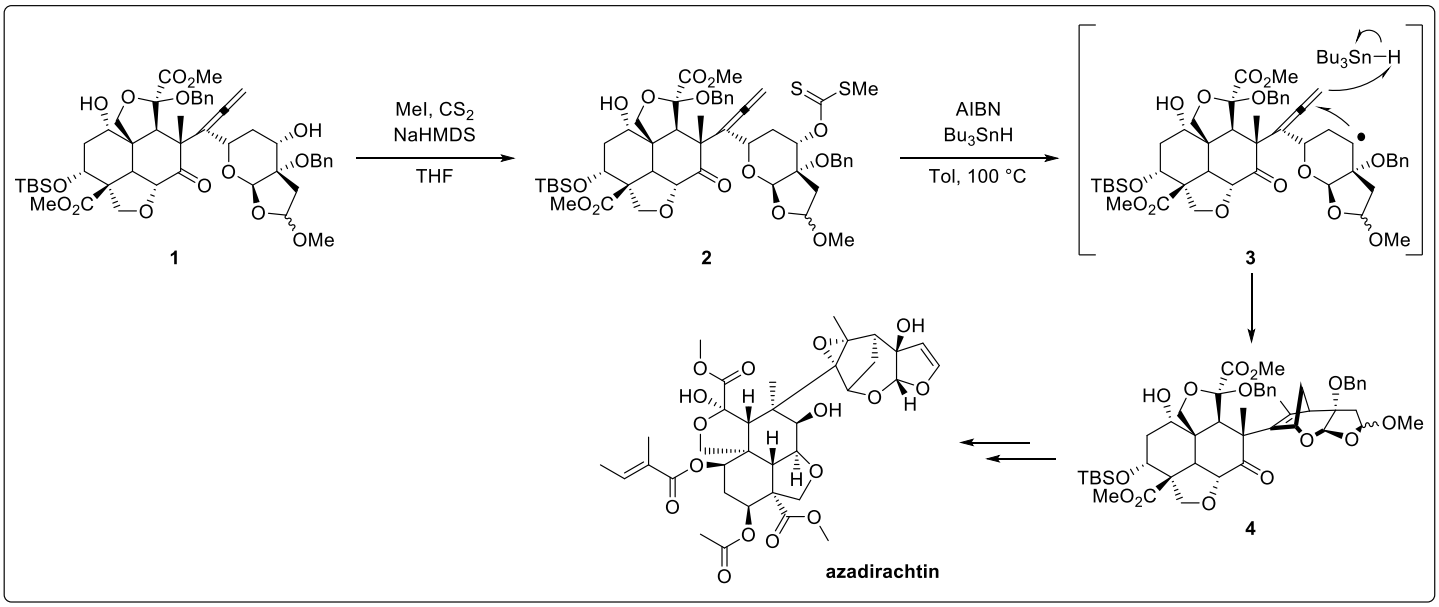
In 1975 Barton and McCombie reported a two-step sequence to replace a hydroxyl-group by a hydrogen atom (*J. Chem. Soc., Perkin Trans.* **1975**, 16, 1574–1585). This powerful radical reaction is applicable to complex structures. The hydroxyl group is first converted to a xanthate group. The radical reactions starts (initiation steps) with the thermal decomposition of AIBN to produce radicals that abstract a hydrogen atom from Bu_3SnH to produce the stannyl radical. The stannyl radical adds to the xanthate producing a new radical species in the first propagation step. In the second propagation step this radical adduct fragments producing an aliphatic radical and a xanthate-stannyl adduct. Finally, hydrogen atom abstraction produces the desired compound and a new stannyl radical. Termination reactions are, for example, the recombination of alkyl and/or stannyl radicals.

4. Propose a product and mechanism for both the reaction with **compound A** and **B** under otherwise identical conditions.



The N-hydroxyphthalimido oxalate, first reported by Okada and co-workers (*J. Am. Chem. Soc.* **1991**, 113, 9401-9402), is photochemically labile (using Ru catalyst and blue LED in this case) thus forming a tertiary radical from the starting material under the reaction conditions reported by Overman and co-workers (*J. Org. Chem.* **2015**, 80, 12, 6025–6036). In both cases the tertiary radical undergoes rapid addition to the **alkene A** or **B**. After addition, in the case of **alkene A**, a hydrogen abstraction from the Hantzsch ester gives the final product. After the addition, in the case of **alkene B**, the labile C-Br bond undergoes β-scission resulting in an alkene product and a bromine atom which then abstracts a hydrogen from the Hantzsch ester.

5. **Compound 1** is an advanced intermediate in the total synthesis of **azadirachtin**. Propose a radical strategy to reach the desired **intermediate 4**. Show the structures of the intermediates and propose reaction conditions.



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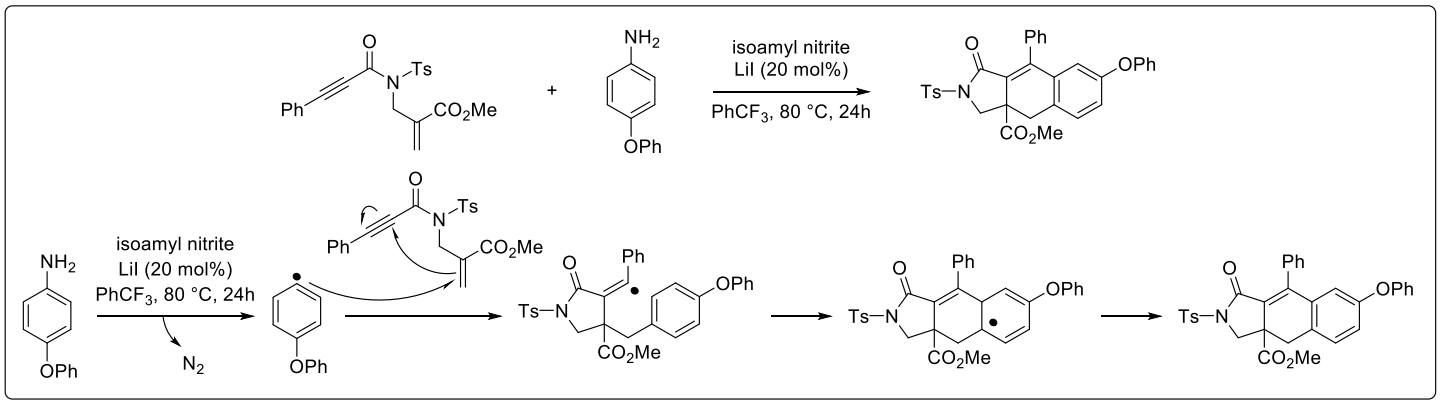
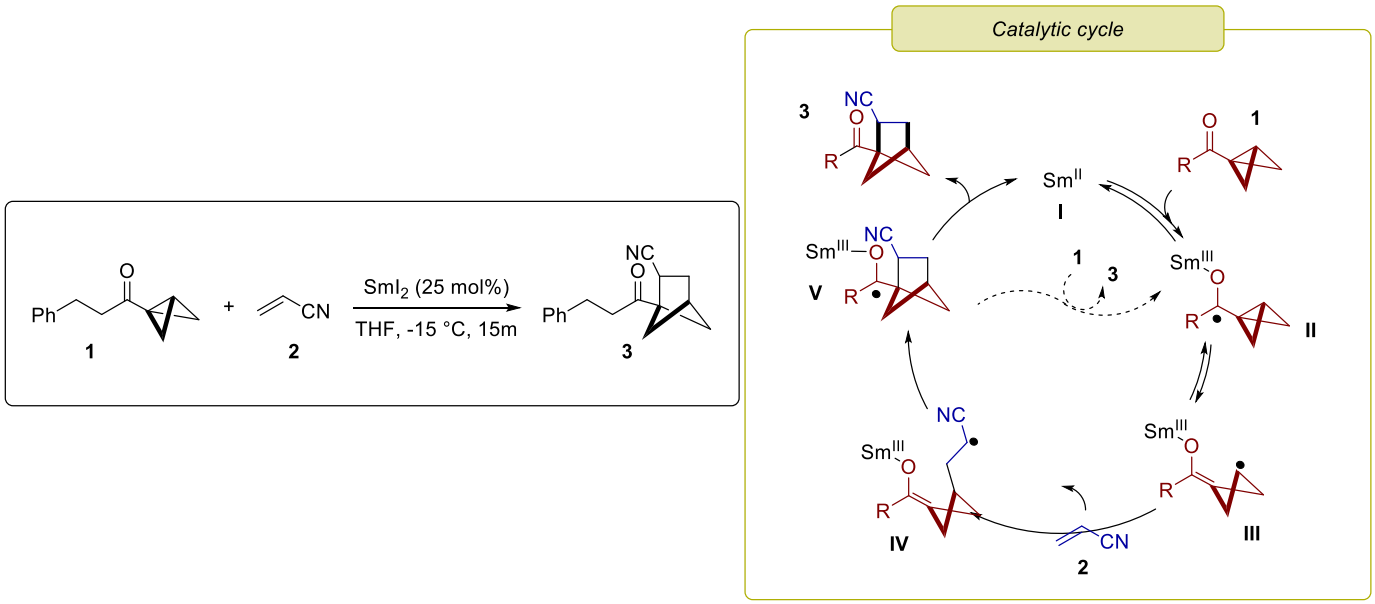
The total synthesis of **azadirachtin** was reported by Ley and co-workers (*Angew. Chem., Int. Ed.* **2007**, 46, 7629-7632). The strategy starts with the conversion of the hydroxyl to a xanthate group under similar conditions as the Barton-McCombie original conditions shown in exercise 3 (in this case NaHMDS instead of NaH was used as base). Barton and McCombie deoxygenation conditions (AIBN and Bu_3SnH as shown in exercise 3) produce the **alkyl radical intermediate 3**. In this case the reaction does not stop and a subsequent 5-exo cyclization of the **intermediate radical 3** onto the allene gives advanced **intermediate 4** which was further converted to the natural product azadirachtin after multiple steps.

6. Propose a product and mechanism for the transformation shown below.



The reaction shown above was reported by Studer and co-workers (*Org. Lett.* **2016**, 18, 6372-6375). The reaction starts with the *in-situ* formation of an aryl radical via the diazonium salt which in its turn is formed from the aniline. The aryl radical is captured by a 1,6-enyne forming an intermediate vinyl radical which ring closes to form a cyclohexadienyl radical. The cyclohexadienyl radical is oxidized to the aromatic product (via SET with the diazonium salt). The authors propose that the Li-salt acts as an initiator through the formation of tiny amounts of a thermally unstable iodoazene (Ar-N=N-I).

7. Propose a catalytic radical cascade mechanism for the transformation shown below.



The Sm-catalyzed intermolecular radical coupling between olefins and bicyclo[1.1.0]butyl ketones was reported by Procter and co-workers (*Nat. Chem.* **2023**, 15, 535-541). The proposed catalytic cycle starts with the formation of the **Sm-ketyl radical II**

which fragments to give **enolate radical III**. Intermolecular coupling with electron deficient **alkene 2** results in **radical IV**. **Radical IV** adds to the Sm(III)-enolate moiety generating a new **ketyl radical V**. Back electron transfer to Sm closes the catalytic cycle and liberates the product. Alternatively, the **ketyl radical V** could directly reduce the starting **ketone 1** (dotted arrows), however this is less likely based on computational calculations.